

LUAD transcriptomic screening report

This is a reproducible *in vitro* expression screen, not a treatment recommendation. The ranked output is in `artifacts/reports/luad_eh3226/all_candidates.csv` after running `scripts/rank_luad_eh3226.py`; input and output hashes are in [data/manifests/eh3226-luad.json](#). The 2.46 GB EH3226 input is fetched separately with `scripts/fetch_eh3226.py`.

Prespecified method and data

GSE32863 contributes 57 verifiable LUAD tumor/normal pairs. The paired exploratory t-test creates the disease log2 fold-change vector. ExperimentHub EH3226 contributes A549 drug perturbation signatures at 10 μ M for 24 hours. On 961 aligned landmark genes, the ranking combines negative Spearman correlation and an up/down gene-set connectivity score using reciprocal-rank fusion ($k=60$). In the aligned landmark set, 57 genes meet the predetermined up threshold and 50 meet the down threshold. No positive-control label or drug-target annotation enters the score.

Top-10 hypotheses

Rank	Drug name in EH3226	Broad sample identity audit
1	hydrocortisone	Same connectivity; different stereochemistry
2	beclomethasone-dipropionate	Exact InChIKey
3	clocortolone-pivalate	Exact InChIKey
4	mometasone	Identity mismatch
5	flumetasone	Ambiguous GEO name
6	BRD-K84203638	No Hub sample
7	hydrocortisone-hemisuccinate	Same connectivity; different stereochemistry
8	fluorometholone	No Hub sample
9	fluticasone	Ambiguous GEO name
10	diflorasone	No Hub sample

The [Broad Drug Repurposing Hub](#) annotates several of these names as steroids or glucocorticoid-receptor agonists. The strong class concentration may reflect a common A549 transcriptional response; it does not establish therapeutic benefit in LUAD. Replaying the upstream first-duplicate selection against official GSE92742 metadata recovers one source `sig_id` and `pert_id` for every Top-10 column; see `configs/luad_top10_signature_ids.csv`. The Broad comparison remains a separate cross-source audit, so an exact source ID does not resolve stereochemistry, annotation mismatches, or absent Hub samples.

Frozen-reference check

Eleven reference names were fixed before ranking. Five exist in the 4,920-name EH3226 A549 subset: docetaxel ranks 106, crizotinib 241, gefitinib 2,444, paclitaxel 2,962, and erlotinib 3,147. The other six are absent and are **unmeasured**, not failures. The spread of these ranks and the top-ranked steroid cluster mean this screen has not demonstrated LUAD drug recovery. No AUC was calculated by treating unmeasured or unknown drugs as negatives.

Evidence and limitations

All ten candidates retain `insufficient_evidence` in their structured ledgers. A frozen [NCBI PubMed E-utilities](#) title/abstract search for each exact drug name with LUAD/NSCLC terms returned 39 hits for hydrocortisone, one for beclomethasone dipropionate, and zero for the other eight names. This narrow query is a retrieval snapshot, not an evidence-absence test; synonyms and broader terms were not searched. Two records were checked at the abstract level and placed in `context_evidence` only: [PMID 35676421](#) independently proposes hydrocortisone as a computational NSCLC candidate, without efficacy validation in its abstract; [PMID 12538830](#) reports glucocorticoid-receptor-dependent CYP3A5 induction by beclomethasone dipropionate in A549 cells, a mechanistic observation rather than anticancer efficacy. Other retrieved hits remain unreviewed.

The expanded [Top-10 identity and evidence matrix](#) resolves the dominant glucocorticoid class, source signature IDs, name-level targets and pathways, candidate-specific context for hydrocortisone and beclomethasone dipropionate, and shared supporting/contradicting evidence. It does not upgrade any candidate: no compound has independently validated LUAD efficacy, dose relevance, safety, or patient-context evidence. The [experimental validation protocol](#) prespecifies dose, molecular-subtype, mechanism, and combination tests, but no wet-lab measurements are claimed. The current expression rank is useful for prioritizing research only.